ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B. Tech. (AI)/ B. Tech. (CC)

Semester: IV

SUBJECT CODE: BCSAI401/BCSI401

DESIGN AND ANALYSIS OF ALGORITHMS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Answer the following questions: (1 × 5 = 5 Marks)

(i) Define algorithms.

(ii) What does it mean for an algorithm to have a linear time complexity (O(n))?

(iii) Explain the Brute-Force method of problem-solving.

(iv) Compare time complexity with space complexity.

(v) Explain the Fractional Knapsack problem.

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) State Master Theorem.

(ii) Distinguish between recursion and iteration.

(iii) What are the primary steps involved in the process of designing an algorithm?

(iv) Explain about Branch and Bound method.

(v) Define Minimum Cost Spanning tree and list its applications.

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) Illustrate greedy algorithm with applications.

OR

(b) Discuss the applications of depth-first search (DFS) and breadth-first search (BFS) in graph traversal. Compare and contrast their time complexities and memory usage.

(ii)(a) Show the result of running the Merge Sorting technique on the sequences 11, 7, 14, 19, 55, 21, 2, 44, 8, 5.

OR

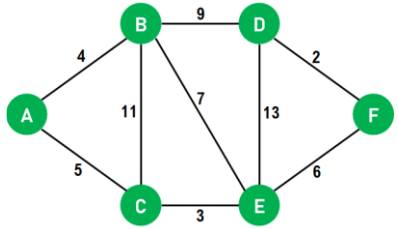
(b) Solve the following instance of the 0/1 knapsack problem using Dynamic programming N=3; (W1, W2, W3) = (23, 15, 7); (P1, P2, P3) = (69, 30, 42); M=20.

Q.4.(i) (a) Insert the following keys in a binary search tree: 5, 12, 77, 23, 44, 50, 3, 86. Then delete the key 77 after adding 209 to the tree.

OR

(b) Define dynamic programming and explain how it differs from greedy algorithms. Provide an example problem (e.g., the longest common subsequence problem) and demonstrate how dynamic programming can be applied to solve it.

(ii)(a) Find shortest path between A and F by using Dijkstra's Algorithm.



OR

(b) Discuss the 4-Queen's problem. Draw the portion of the state space tree for n = 4 queens using the backtracking algorithm.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5.(a) Explain P, NP, NP-complete and NP-hard.

OR

(b) Explain the DFS algorithm. Draw a neat labelled diagram to show each step.

Q.6.(a) Use the Master theorem to solve the recurrence relation T(n)=2T(n/2)+n2. Determine the time complexity of the algorithm and justify your solution.

OR

(b) Discuss Randomized algorithm with example.